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ORIGINAL

Characteristics of ventilator weaning in patients intubated for Guillain–Barré syndrome or myasthenia gravis: a nationwide multicenter study

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Abstract

Purpose: Severe forms of Guillain–Barré syndrome (GBS) and myasthenia gravis (MG) may result in prolonged mechanical ventilation in ICUs. Data on weaning from mechanical ventilation are scarce, and our objective was to assess ventilator weaning characteristics in a large-scale multicenter study.

Methods: Multicenter retrospective cohort study including all patients intubated for at least 48 h for GBS or MG over a 10-year period (from January 2014 to December 2023). The primary outcome was the proportion of patients having experienced prolonged weaning (at least 7 days).

Results: Some 886 patients admitted to 47 ICUs in France were retained in the analysis, including 513 (58%) with GBS and 373 (42%) with MG. Patients with GBS were more likely to experience prolonged weaning than those with MG (64% vs. 35%, $p < 0.001$). An extubation attempt was performed in only 46% of patients with GBS and in 88% patients with MG ($p < 0.001$). Reintubation rates were 26% in patients with GBS and 29% in patients with MG ($p = 0.440$). Patients with GBS were more likely to undergo tracheostomy than those with MG (57% vs. 17%, $p < 0.001$). Patients with GBS required a longer duration of mechanical ventilation and ICU stay. Mortality rates were similar (9.4% for patients with GBS and 9.7% for patients with MG, $p = 0.882$).

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Conclusions: Patients with GBS were more likely to experience prolonged weaning and to undergo tracheostomy than patients with MG. More than one-fourth of patients required reintubation after an extubation attempt. Despite prolonged duration of mechanical ventilation, mortality remained relatively low.

Keywords: Ventilator weaning, Airway extubation, Tracheostomy, Guillain–Barré syndrome, Myasthenia gravis, Intensive care unit

Introduction

Severe forms of both Guillain–Barré syndrome (GBS) and myasthenia gravis (MG) may lead to acute respiratory failure and a need for mechanical ventilation [1–4]. In these patients, respiratory failure is primarily due to respiratory muscle weakness and to swallowing disorders that potentially result in aspiration pneumonia. Both diseases require prolonged duration of mechanical ventilation in ICUs and frequently present ventilator weaning difficulties. The need for tracheostomy is common while awaiting neuromuscular recovery sufficient to allow physiological spontaneous ventilation, as well as the restoration of effective cough and swallowing capacity [5–12]. However, data on weaning from mechanical ventilation are scarce in these patients and primarily come from single-center or small-scale multicenter studies [5–13]. Although prolonged spontaneous breathing trials have recently been suggested as a way to prevent extubation failure especially in patients with MG which is characterized by respiratory muscle fatigue, these interesting findings were observed in a small sample size at a single center [13]. A systematic review revealed a lack of studies showing the benefits of any specific weaning protocol in mechanically ventilated patients with neuromuscular disease [14]. Data on weaning from mechanical ventilation in patients with GBS are even scarcer than data on patients with MG, and a recent narrative review on GBS acknowledged that due to the limited number of small-scale, often single-center, retrospective studies, weaning modalities were primarily based on expert opinions [5]. To date, no study has compared ventilator weaning modalities in patients with these two neuromuscular diseases. Although GBS and MG are two distinct acquired neuromuscular diseases, both are commonly managed in ICUs and present similar weaning difficulties, with particularly prolonged weaning and frequent need for tracheostomy. However, GBS and MG may require different ventilator weaning protocols and comparing the two diseases could highlight their differences. Our objectives were to assess ventilator weaning difficulties and to compare weaning characteristics between patients mechanically ventilated for GBS or for MG in a large-scale multicenter study.

Take-home message:

Patients with Guillain-Barre syndrome are more likely to experience prolonged weaning and to undergo tracheostomy than patients with myasthenia gravis. More than one-fourth of patients required reintubation after an extubation attempt. Despite prolonged duration of mechanical ventilation, mortality remained relatively low.

Methods

Design and patients

Multicenter retrospective cohort study including all patients mechanically ventilated for at least 48 h due to GBS or MG over a 10-year period (from January 2014 to December 2023) in French ICUs. Medical records were identified using their International Classification of Diseases 10th revision (G610 for GBS and G700 for MG). Each investigator was responsible for selecting patients with the correct diagnosis and for excluding patients who had been intubated for less than 48 h or for reasons other than neuromuscular disease. This study was approved by the independent institutional review board of Poitiers, and registered at http://www.clinicaltrials.gov (NCT07022028). In compliance with the national legislation regarding observational retrospective studies, this study was declared to European General Data Protection Regulation (MR004). An information letter was sent to all patients eligible in the study, and patients could be included on the condition that they did not object.

Weaning difficulty

Weaning was classified as short, intermediate, or prolonged according to the most recent definitions [15, 16]. *Short weaning* was defined as complete successful liberation from the ventilator less than 24 h after the initial spontaneous breathing trial (SBT). *Intermediate weaning* was defined as complete successful liberation from the ventilator more than 24 h and less than 7 days after the initial SBT. *Prolonged weaning* was defined as still being under mechanical ventilation at least 7 days after the initial SBT. In case of extubation without prior SBT (self-extubation or decision of extubation without prior SBT), the date of extubation was considered as being the initial attempt at separation from the ventilator.

For patients having undergone tracheostomy without prior SBT, we considered the initial separation attempt from the ventilator via tracheostomy to be the start of ventilator weaning. Tracheostomy could be performed according to the following three strategies: (1) direct tracheostomy without prior SBT, (2) after SBT failure without prior extubation attempt, or (3) after extubation failure leading to reintubation.

Patients who died without any prior weaning attempt (neither SBT, nor extubation or attempt at separation from the ventilator with tracheostomy) were classified as "no weaning process." Patients who died within the first 7 days following the initial SBT and who therefore could not be classified according to weaning difficulty were considered as “weaning failure.”

Outcomes

The primary outcome was the proportion of patients with prolonged weaning defined as still being under mechanical ventilation at least 7 days after the initial attempt at separation from the ventilator (i.e., after the initial SBT, after the initial extubation attempt in case of extubation without prior SBT, or after the initial separation attempt from the ventilator with tracheostomy in patients with direct tracheostomy).

Main secondary outcomes included weaning modalities, *i.e.,* the proportion of patients having undergone SBT, an extubation attempt, and a tracheostomy. The other secondary outcomes included need for reintubation, duration of mechanical ventilation, length of stay in ICU, and survival or death in ICU.

Sample size calculation

Based on previous studies reporting prolonged weaning in approximately half of patients with MG and two-thirds of patients with GBS [7, 11–13], we calculated than enrollment of around 800 patients would provide the study with 80% power to show an absolute difference of 10% in the rate of prolonged weaning between patients with MG and those with GBS (50% and 60%, respectively), at a two-sided alpha level of 0.05.

We estimated a potential for inclusion of around two patients per year and per center, *i.e.,* 20 patients per center over the planned 10-year period of inclusion. Therefore, we needed at least 40 participating centers. To be certain to include a sufficient number of patients, 50 centers were solicited in France to be part of the study.

Statistical analysis

All of the analyses were performed by the study statistician (L.C.). Continuous variables are expressed as mean ± standard deviation or median [interquartile range, 25th–75th percentiles], and qualitative variables as number and percentage.

The primary outcome (prolonged weaning) was compared between patients with GBS and those with MG by means of the χ^2 test. Statistical comparisons of characteristics and outcomes were performed between three sets of patient groups: (1) according to the type of neuromuscular disease (GBS *versus* MG), (2) according to the weaning difficulty (prolonged weaning versus intermediate or short weaning), and (3) according to the extubation outcome (reintubation versus successful extubation). For categorical variables, χ^2 tests or Fisher exact test were used as appropriate, while continuous variables were analyzed using Student’s t-test or Welch test the Wilcoxon test, depending on data distribution. A multivariable logistic regression analysis was performed to identify variables independently associated with prolonged weaning. Variables associated with a *p* value less than 0.20 after univariable analysis were entered into the maximal model. Kaplan–Meier curves were plotted in patients with GBS and MG to assess the probability of reintubation during the 14 days following extubation and compared by means of a log-rank test. Statistical analyses were conducted by using R software version 4.0.4 (R Foundation for Statistical Computing; https://www.R-project.org). A two-tailed *p* value of less than 0.05 was considered statistically significant.

Results

Some 903 patients admitted to 47 ICUs in France were eligible for inclusion. After excluding 17 patients due to missing data regarding the primary outcome, 886 patients were retained in the analysis including 513 (58%) with GBS and 373 (42%) with MG. The diagnosis was made in the ICU in 59% of GBS cases (using electromyography in 94% of cases, lumbar puncture in 88%, and antibodies in 30%), and in 19% of MG cases (using electromyography in 70% of cases and antibodies in 75%). Among the 513 patients with GBS, 221 patients (43%) experienced bulbar weakness, and 123 (24%) had anti-ganglioside antibodies. Their treatment regimen included intravenous immunoglobulins or therapeutic plasma exchange in 98% of cases (504 patients). Among the 373 patients with MG, 261 (72%) had acetylcholine receptor antibodies, 18 (5%) had anti-MuSK, and 119 (32%) had a medical history of thymoma. Their treatment regimens included cholinesterase inhibitors in 95% of cases (355 patients), intravenous immunoglobulins or therapeutic plasma exchange in 88% (327 patients), steroids in 80% (300 patients), and other immunosuppressive drugs in 38% (143 patients). Patients with MG were older and had a higher severity score at admission than patients with GBS (Table 1). They were also more likely to have underlying chronic cardiac and restrictive pulmonary diseases. Patients with GBS required a longer duration of mechanical ventilation and a longer ICU stay than those with MG. Mortality rates in ICU were similar (MG 9.7% *vs.* GBS 9.4%, $p=0.882$).

Weaning modalities in the overall population

Among the 886 patients studied, 198 (22%) had a short weaning, 189 (21%) had an intermediate weaning, 460 (52%) had prolonged weaning, seven (1%) died within the first 7 days following weaning initiation (weaning failure), and 32 (4%) died with no weaning process. Males were more likely to experience prolonged weaning than females (Table 2), even after adjustment on the neuromuscular disease (aOR 1.41; 95% CI [1.06 to 1.88], $p=0.017$). Extubation was attempted at least once in 566 patients (64%). Among them, extubation was planned after a prior SBT in 94% of cases (530 patients), planned without a prior SBT in 2% (12 patients), and unplanned after self-extubation in 4% (24 patients). The overall reintubation rate after extubation was 28% (156 out of 566 patients). Tracheostomy was performed in 354 patients (40%). All of them had prolonged weaning. Changes in weaning characteristics over time are given in Table S1.

Comparison of weaning modalities between patients with MG and GBS

Regarding the primary outcome, patients with GBS were more likely to experience prolonged weaning than those with MG (64% vs. 35%, $p<0.001$) (Fig. 1), even after adjustment on sex (aOR 3.43; 95% CI [2.58 to 4.58], $p<0.001$). In contrast, patients with MG were more likely to experience short and intermediate weaning, and to undergo an extubation attempt (MG 88% vs. GBS 46%, $p<0.001$) (Table 3). After extubation, the reintubation rate was 29% for patients with MG and 26% for patients with GBS ($p=0.440$) (Fig. 2). Patients with GBS were more likely to undergo tracheostomy (57% vs. 17%, $p<0.001$), often directly without any prior SBT than in patients with MG (70% vs. 30%, $p<0.001$). Patients with GBS were more likely to undergo tracheostomy decannulation before ICU discharge than those with MG (56% vs. 41%, p = 0.031) (Table 3).

Factors associated with reintubation

Among the 530 patients extubated after a prior SBT, reintubation rates in ICU were 28% (89 out of 315 patients) in patients with MG and 25% (53 out of 215 patients) in patients with GBS ($p=0.357$). Patients who required reintubation were older and had higher severity scores at ICU admission than those successfully extubated (Table 4). Patients who required reintubation underwent

Table 1 Comparison of patients with Guillain–Barré syndrome and myasthenia gravis

	Guillain–Barré (*n* = 513)	Myasthenia gravis (*n* = 373)	*P* value
Characteristics of the patients			
Age, years	57 ± 18	65 ± 17	**< 0.001**
Male sex, *n* (%)	295 (57%)	198 (53%)	0.211
Body mass index, kg/m^2	26 ± 6	25 ± 5	0.168
SAPS II at admission, points	33 ± 15	40 ± 19	**< 0.001**
Underlying chronic cardiac disease, *n* (%)	47 (9%)	58 (16%)	**0.004**
Left ventricular dysfunction, *n* (%)	6 (1%)	13 (3%)	**0.019**
Ischemic heart disease, *n* (%)	27 (5%)	28 (8%)	0.172
Atrial fibrillation, *n* (%)	22 (4%)	27 (7%)	0.058
History of cardiogenic pulmonary edema, *n* (%)	5 (1%)	9 (2%)	0.090
Underlying chronic respiratory disease, *n* (%)	39 (8%)	41 (11%)	0.082
Chronic obstructive pulmonary disease, *n* (%)	31 (6%)	26 (7%)	0.578
Obesity hypoventilation syndrome, *n* (%)	7 (1%)	6 (2%)	0.765
Chronic restrictive pulmonary disease, *n* (%)	2 (0.4%)	17 (5%)	**< 0.001**
Outcomes			
Total duration of mechanical ventilation, days	25 [13–51]	9 [6–21]	**< 0.001**
ICU length of stay, days	35 [19–65]	15 [9–28]	**< 0.001**
Death in ICU, *n* (%)	48 (9.4%)	36 (9.7%)	0.882

Values are given in mean ± standard deviation or median [interquartile range, 25–75 percentiles] and number (%)

SAPS II Simplified Acute Physiology Score, *ICU* Intensive Care Unit

In bold are indicated *p* values considered as statistically significant ($p<0.05$)

Table 2 Comparison between patients who experienced prolonged weaning (at least 7 days) and those with short or intermediate weaning

	Prolonged ($n = 460$)	Short or Intermediate ($n = 387$)	p value
Characteristics of the patients			
Neuromuscular disease			**< 0.001**
Guillain–Barré syndrome	329 (72%)	163 (42%)	
Myasthenia gravis	131 (28%)	224 (58%)	
Age, years	60 ± 17	60 ± 18	0.960
Male sex, *n* (%)	273 (59%)	195 (50%)	**0.009**
Body mass index, kg/m^2	26 ± 5	26 ± 6	0.974
SAPS II at admission, points	35 ± 17	35 ± 17	0.646
Underlying chronic cardiac disease, *n* (%)	48 (10%)	50 (13%)	0.260
Left ventricular dysfunction, *n* (%)	8 (12%)	9 (2%)	0.544
Ischemic heart disease, *n* (%)	25 (5%)	29 (7%)	0.222
Atrial fibrillation, *n* (%)	28 (6%)	17 (4%)	0.273
History of cardiogenic pulmonary edema, *n* (%)	6 (1%)	7 (2%)	0.552
Underlying chronic respiratory disease, *n* (%)	37 (8%)	35 (9%)	0.603
Chronic obstructive pulmonary disease, *n* (%)	27 (6%)	25 (6%)	0.721
Obesity hypoventilation syndrome, *n* (%)	5 (1%)	7 (2%)	0.376
Chronic restrictive pulmonary disease, *n* (%)	8 (2%)	8 (2%)	0.727
Weaning characteristics			
At least one SBT performed, *n* (%)	206 (45%)	365 (94%)	**< 0.001**
Type of the initial SBT, *n* (%)			0.746
T-piece, *n* (%)	127 (62%)	220 (60%)	
Pressure support ventilation, *n* (%)	79 (38%)	145 (40%)	
Duration of the initial SBT, min	60 [30–90]	60 [45–120]	**< 0.001**
Success of the initial SBT, *n* (%)	111 (54%)	282 (77%)	**< 0.001**
Total number of SBTs performed, *n*	3 [1–6]	1 [1–3]	**< 0.001**
Maximal duration of SBTs, min	180 [80–480]	120 [60–270]	**< 0.001**
At least one prolonged SBT lasting > 2h, *n* (%)	109 (53%)	136 (37%)	**< 0.001**
At least one prolonged SBT lasting ≥ 8h, *n* (%)	48 (23%)	39 (11%)	**< 0.001**
At least one extubation attempt, *n* (%)	175 (38%)	387 (100%)	**< 0.001**
Prophylactic NIV after extubation, *n* (%)	69/175 (39%)	102/387 (26%)	**0.002**
High-flow nasal oxygen after extubation, *n* (%)	17/175 (10%)	27/387 (7%)	0.263
Tracheostomy performed, *n* (%)	354 (77%)	0 (0%)	**< 0.001**
Outcomes			
Total duration of mechanical ventilation, days	36 [21–62]	8 [5–13]	**< 0.001**
ICU length of stay, days	46 [29–83]	12 [9–19]	**< 0.001**
Death in ICU, *n* (%)	41 (9%)	4 (1%)	**< 0.001**

Values are given in mean ± standard deviation or median [interquartile range, 25–75 percentiles] and number (%)

SAPS II Simplified Acute Physiology Score, *ICU* Intensive Care Unit

In bold are indicated p values considered as statistically significant ($p < 0.05$)

their initial SBT at an earlier time than those who were successfully extubated. At the time of extubation, ventilator settings and blood gases did not significantly differ between patients who required reintubation and those who were successfully extubated, except for larger tidal volumes in the latter. Patients who were successfully extubated were also more likely to have effective cough than those who required reintubation. The rate of reintubation reached 61% in patients extubated with ineffective cough (40 out of 66 patients). Patients who required reintubation had worse outcomes including longer duration of mechanical ventilation, longer ICU stay, and higher mortality in ICUs (Table 4).

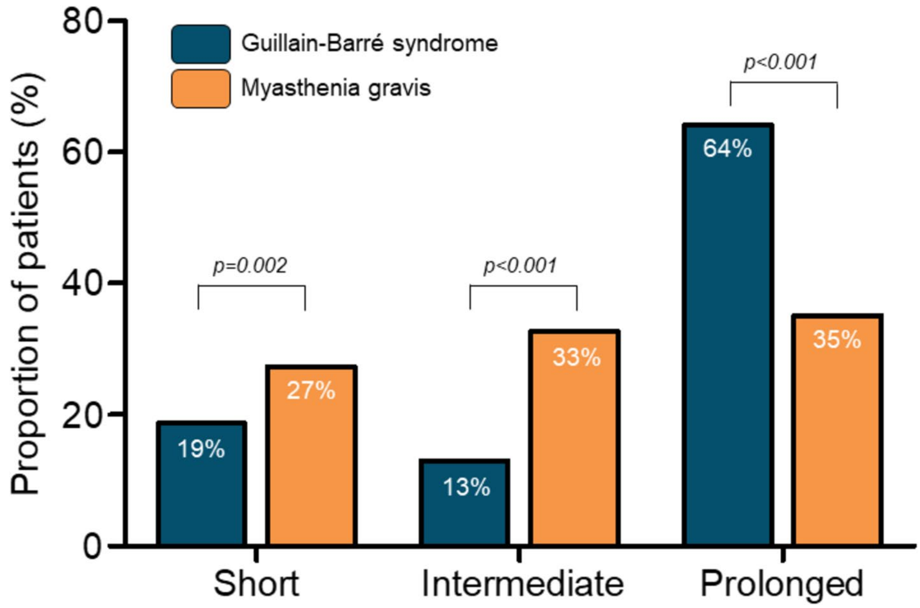

Fig. 1 Histograms showing the proportion of patients with short weaning (complete successful liberation from the ventilator less than 24 h after the initial attempt at separation from the ventilator), intermediate weaning (more than 24h and less than 7 days), and prolonged weaning (still under mechanical ventilation at least 7 days after the initial attempt at separation from the ventilator). Patients with Guillain–Barré syndrome (GBS represented by blue histograms) were more likely to experience prolonged weaning than patients with myasthenia gravis (MG represented by orange histograms) (64% vs. 35%, $p < 0.001$)

Discussion

This large-scale multicenter cohort demonstrates that during the study period, weaning from mechanical ventilation was markedly more prolonged in patients with GBS than in those with MG. Most patients with GBS underwent tracheostomy, which was most often performed directly without prior SBT. Consequently, less than half of patients with GBS underwent at least one extubation attempt, which was the case in nearly 90% of the patients with MG. After an extubation attempt, reintubation was required in 25–30% of patients, with no significant difference in reintubation rates between patients with GBS and those with MG.

Prolonged weaning

Overall, only 22% of patients experienced short weaning. This is rather the opposite than in the general ICU population, where short weaning occurs in 75–80% of patients, while prolonged weaning affects no more than 10% [15–18]. Males were more likely than females to experience prolonged weaning. It is unclear whether this is related to pathophysiological differences or to differences in medical management related to sex. A recent study suggested that compared to females, males may be at increased risk of reintubation, but the reasons remain unclear [19], and we believe that a sex gap should be systematically examined in the future.

Herein, weaning was prolonged in approximately one-third of patients with MG and two-thirds of patients with GBS. These rates are somewhat lower than those reported in literature, which range from 40 to 60% in patients with MG [11–13], and from 60% to over 90% in patients with GBS [6–9]. However, most previous studies have reported rates of weaning difficulties including reintubation or tracheostomy, without specifying weaning duration and rates of prolonged weaning [6–12]. We report weaning difficulties according to the most recent classification on weaning, which enables identification of patients with a weaning prolonged for at least 7 days [15, 16]. These definitions have the advantage of allowing the classification of all intubated patients, even those who died with no weaning process, and therefore of providing accurate and comprehensive rates for the entire population.

Risk of reintubation

We observed similar reintubation rates in patients with MG or GBS (29% and 26%, respectively). These rates are in keeping with that reported with literature, ranging from 14 to 43% in patients with MG [11–13], and of 30% in one of the only studies reporting reintubation rates in patients with GBS [7]. At the time of extubation, patients who required reintubation had less cough strength and lower tidal volumes than patients who were successfully

Table 3 Comparison of the weaning process between patients with Guillain–Barré syndrome and those with myasthenia gravis

	Guillain–Barré (*n* = 513)	Myasthenia gravis (*n* = 373)	*p* value
Weaning classification, *n* (%)			
Short weaning, *n* (%)	96 (19%)	102 (27%)	**0.002**
Intermediate weaning, *n* (%)	67 (13%)	122 (33%)	**< 0.001**
Prolonged weaning, *n* (%)	329 (64%)	131 (35%)	**< 0.001**
No weaning or weaning failure, *n* (%)	21 (4%)	18 (5%)	0.600
Spontaneous breathing trials			
At least one SBT performed, *n* (%)	252 (49%)	325 (87%)	**< 0.001**
Time between intubation and the initial SBT, days	8 [4–13]	4 [2–7]	**< 0.001**
Type of the initial SBT, *n* (%)			0.623
T-piece, *n* (%)	150 (60%)	200 (54%)	–
Pressure support ventilation, *n* (%)	102 (40%)	125 (46%)	–
Duration of the initial SBT, min	60 [30–100]	60 [45–120]	**0.004**
Success of the initial SBT, *n* (%)	177 (70%)	218 (67%)	0.418
Total number of SBTs performed, *n*	1 [1–3]	2 [1–4]	0.084
Maximal duration of SBTs, min	120 [60–240]	135 [70–390]	**< 0.001**
At least one prolonged SBT lasting > 2h, *n* (%)	95 (38%)	152 (47%)	**0.029**
At least one prolonged SBT lasting ≥ 8h, *n* (%)	23 (9%)	62 (19%)	**0.001**
Extubation			
At least one extubation attempt, *n* (%)	236 (46%)	330 (88%)	**< 0.001**
Time between intubation and first extubation, days	10 [5–16]	6 [4–11]	**< 0.001**
Prophylactic NIV after extubation, *n* (%)	62 (26%)	110 (33%)	0.072
High-flow nasal oxygen after extubation, *n* (%)	19 (8%)	25 (8%)	0.835
Reintubation, *n* (%)	61 (26%)	95 (29%)	0.440
Mortality of reintubated patients, *n* (%)	4/61 (7%)	15/95 (16%)	0.085
Tracheostomy			
Tracheostomy performed, *n* (%)	290 (57%)	64 (17%)	**< 0.001**
Time between intubation and tracheostomy, days	12 [6–19]	18 [11–26]	**< 0.001**
Type of tracheostomy			**< 0.001**
Direct tracheostomy, *n* (%)	222 (77%)	19 (30%)	
After SBT failure, *n* (%)	33 (11%)	9 (14%)	
After extubation failure, *n* (%)	35 (12%)	36 (56%)	
Deventilation before ICU discharge, *n* (%)	220 (76%)	50 (78%)	0.700
Decannulation before ICU discharge, *n* (%)	161 (56%)	26 (41%)	**0.031**

Values are given in mean ± standard deviation or median [interquartile range, 25–75 percentiles] and number (%)

SBT Spontaneous Breathing Trial, *NIV* Non-Invasive Ventilation, *ICU* Intensive Care Unit

In bold are indicated *p* values considered as statistically significant ($p < 0.05$)

extubated. In the general ICU population, cough strength is a major risk factor for reintubation, with reintubation rates exceeding 30% to 35% in patients who are extubated with an ineffective cough [20–23]. Indeed, the reintubation rate was over 60% for patients with GBS or MG who were extubated with an ineffective cough. Spontaneous breathing trials alone are insufficient to assess readiness for extubation. Recovery of an effective cough may be an even more critical factor in the decision to extubate patients with neuromuscular diseases. Accurate measurements of cough strength using peak flow [20], and the effectiveness of prophylactic measures after extubation on extubation success (non-invasive ventilation, cough augmentation techniques, physiotherapy…) should be specifically investigated in this population [24–26].

Although the reported values were not discriminatory, lower tidal volumes could be a reflection of lower vital capacity, insofar as has been suggested that they may predict respiratory dysfunction in patients with MG or GBS [27, 28]. Moreover, patients who required reintubation had been extubated few days earlier than those successfully extubated. Higher reintubation rates

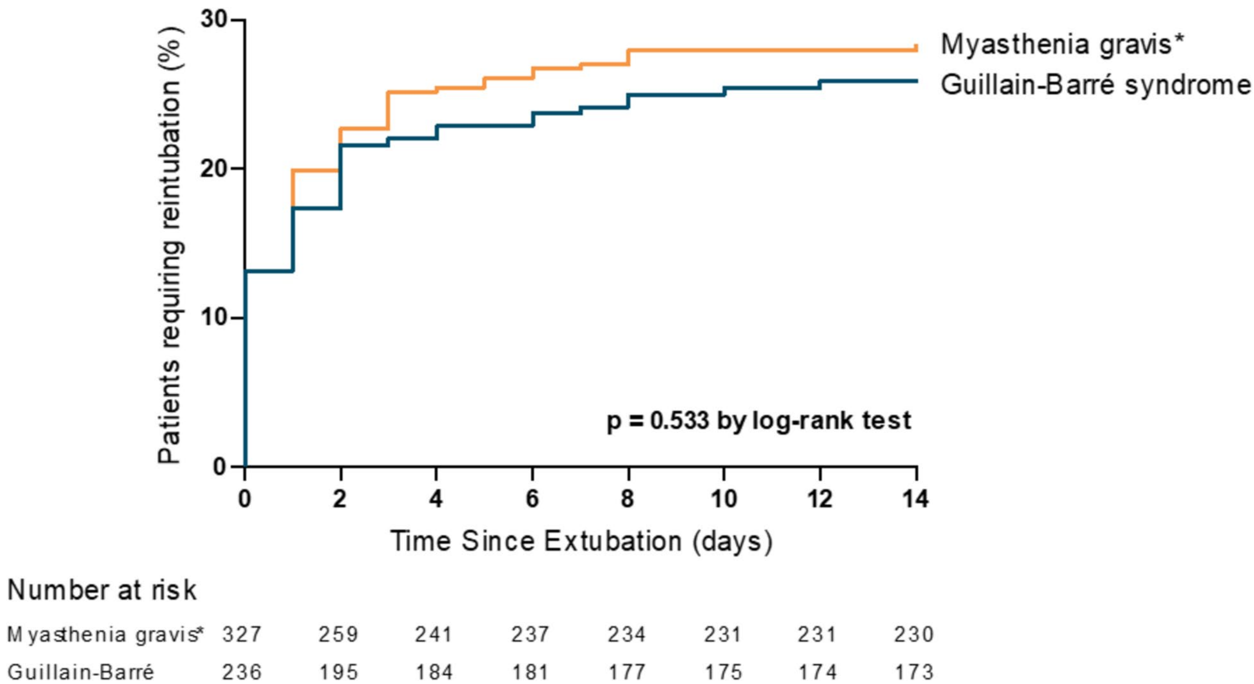

Fig. 2 Kaplan–Meier curves showing that the probability of reintubation during the 14 days following extubation did not significantly differ between patients with Guillain–Barré syndrome (GBS) in blue and those with myasthenia gravis (MG) in orange ($p=0.533$ using log-rank test). We selected a time frame of 14 days because all reintubations occurred within the first 2 weeks following extubation. The rate of reintubation in ICU was 26% in patients with GBS and 29% in those with MG ($p=0.440$ using the χ^2 test). *Three patients who required reintubation (all with myasthenia gravis) were excluded from Kaplan–Meier curves because the date of reintubation could not be specified

were previously reported in patients with MG having undergone extubation during the first week following intubation, as compared to patients extubated within the second week [12]. This is unlike the general ICU population, in which extubation after the first successful SBT is associated with decreased mechanical ventilation duration without an increased risk of reintubation [29, 30]. In patients with GBS or MG, prompt extubation after a short duration of mechanical ventilation could be associated with an increased risk of reintubation that may be due to incomplete recovery of cough strength and bulbar functions. A recent study showed that SBTs performed over increasingly long periods until the patient was able to breathe spontaneously over an 8-h period were associated with particularly low reintubation rates in patients with MG [13]. By contrast, we observed higher reintubation rates in patients having undergone prolonged SBTs of at least eight hours. Therefore, prolonged SBTs do not appear to reduce the risk of reintubation. One hypothesis to explain this difference is that extubation after repeated successful SBTs while waiting for disease improvement, especially recovery of cough strength and bulbar functions, may be more important than the actual duration of the SBT. Furthermore, the aforementioned study showed that most reintubations occurred due to insufficient stabilization of the neuromuscular disease [13].

Tracheostomy

Even though the proportion of patients having undergone tracheostomy was particularly high in patients with GBS, this rate is lower than in the literature, which exhibits ranges from 57% (exactly as in our study) to more than 90% of cases [6–9]. The timing of tracheostomy had never been specified in previous studies, and only one study specified that tracheostomy was directly performed without prior SBT in 80% of cases [7], *i.e.*, very close to our 77% rate. Regarding patients with MG, two studies reported rates of tracheostomy ranging from 25 to 30%, which is also slightly higher than the 17% rate observed in our study. Tracheostomy was performed earlier in patients with GBS, which may suggest that clinicians expected a longer recovery course than in patients with MG.

Limitations

The main limitation is obviously the retrospective nature of our study, with inherent information bias. However, this is the largest study reporting on weaning

Table 4 Comparison between patients who required reintubation and those who were successfully extubated among all patients extubated after a prior spontaneous breathing trial

	Reintubation ($n = 142$)	No reintubation ($n = 388$)	p value
Characteristics of the patients			
Age, years	65 ± 17	59 ± 18	**0.001**
Male sex, *n* (%)	83 (58%)	190 (49%)	0.053
Body mass index, kg/m^2	25 ± 5	25 ± 6	0.217
SAPS II at admission, points	42 ± 18	35 ± 18	**< 0.001**
Underlying chronic cardiac disease, *n* (%)	20 (14%)	50 (13%)	0.718
Underlying chronic respiratory disease, *n* (%)	13 (9%)	33 (9%)	0.814
Characteristics at time of extubation			
Pressure support ventilation mode, *n* (%)	129 (91%)	347 (89%)	0.634
Pressure support level, cm H_2O	8 [7–10]	8 [7–10]	0.114
PEEP level, cm H_2O	5 [5, 6]	5 [5, 6]	0.899
RR, breaths per minute	21 [16–25]	20 [16–23]	0.058
Tidal volume, ml	447 [368–540]	450 [400–550]	0.113
Tidal volume, ml/kg of predicted body weight	6.9 [5.6–8.0]	7.3 [6.3–8.9]	**0.007**
pH, units	7.45 [7.42–7.48]	7.46 [7.43–7.49]	0.130
$PaCO_2$, mm Hg	40 [35–43]	39 [36–42]	0.535
PaO_2/FiO_2, mm Hg	297 [255–400]	318 [248–381]	0.851
Effective cough*, *n* (%)	35/101 (35%)	260/334 (78%)	**< 0.001**
Ineffective cough*, *n* (%)	40/101 (40%)	26/334 (8%)	**< 0.001**
Weaning characteristics			
Time between intubation and the initial SBT, days	4 [2–8]	5 [3–11]	**0.041**
Time between intubation and first extubation, days	6 [3–12]	8 [5–14]	**0.014**
Type of the initial SBT, *n* (%)			0.728
T-piece, *n* (%)	88 (62%)	234 (60%)	
Pressure support ventilation, *n* (%)	54 (48%)	154 (40%)	
Duration of the initial SBT, min	60 [34–89]	60 [42–120]	0.070
Success of the initial SBT, *n* (%)	101 (71%)	278 (72%)	0.906
Total number of SBTs performed, *n*	2 [1–3]	2 [1–3]	0.637
Maximal duration of SBTs, min	150 [60–390]	120 [60–330]	0.210
At least one prolonged SBT lasting > 2 h, *n* (%)	65 (46%)	159 (41%)	0.322
At least one prolonged SBT lasting ≥ 8 h, *n* (%)	29 (20%)	46 (12%)	**0.012**
Prophylactic NIV after extubation, *n* (%)	47 (33%)	115 (30%)	0.444
High-flow nasal oxygen after extubation, *n* (%)	16 (11%)	25 (6%)	0.066
Outcomes			
Total duration of mechanical ventilation, days	27 [15–40]	9 [6–15]	**< 0.001**
ICU length of stay, days	36 [23–54]	13 [9–21]	**< 0.001**
Death in ICU	19 (13%)	6 (2%)	**< 0.001**

Values are given in mean ± standard deviation or median [interquartile range, 25–75 percentiles] and number (%)

SAPS II Simplified Acute Physiology Score, *PEEP* Positive End-Expiratory Pressure, *SBT* Spontaneous Breathing Trial, *NIV* Non-Invasive Ventilation, *ICU* Intensive Care Unit

* Cough strength was assessed using a semi-quantitative 5-point Likert scale, which was reported by nurses or physiotherapists or obtained from medical records. Cough strength was graded as follows: 0 (absent), 1 (weak), 2 (moderate), 3 (effective), or 4 (strong). It was then defined as "ineffective" for scores of 0 or 1, neutral for a score of 2, and "effective" for scores of 3 or 4

In bold are indicated *p* values considered as statistically significant ($p < 0.05$)

modalities in these two neuromuscular diseases. This is a comprehensive general description classifying a large number of patients admitted to numerous units (teaching as well non-teaching hospitals), using the most recent classification on weaning [15, 16]. Therefore, we comprehensively indicated all weaning difficulties, including not only prolonged weaning and reintubation rates but also the need for tracheostomy, of which the timing and the proportion of patients having undergone direct tracheostomy was clearly specified. Another major limitation is the lack of follow-up beyond ICU discharge and especially until complete decannulation in weaning centers. However, patients discharged with tracheostomy still in place represented less than 20% (167 patients) of the whole population. The lack of data on the intensity of muscle weakness prevents us from assessing the impact of disease severity on weaning difficulties or the risk of extubation failure. However, we included a relatively homogeneous subset of patients with particularly severe cases, of whom were intubated in ICUs. Lastly, we conducted a large number of exploratory statistical comparisons, increasing the risk of a type I error. This means that some significant results could be false positives and should be interpreted cautiously.

Conclusion

Patients with GBS were more likely than those with MG to experience prolonged weaning. Males were also more likely than females to experience prolonged weaning. Most patients with GBS were weaned with a tracheostomy, and in more than three-fourths of the cases, the tracheostomy was performed directly without a prior SBT or attempt at extubation. More than one-fourth of patients required reintubation after an extubation attempt. These rates were comparable between patients with GBS or MG. However, early extubation seemed to be associated with an increased risk of reintubation. Despite prolonged duration of mechanical ventilation, mortality remained relatively low.

Supplementary Information

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Author contributions

AWT and LC had full access to all of the data in the study. AWT and SLP take responsibility for the integrity of the data and the accuracy of the data analysis. AWT designed the study. AWT wrote the manuscript. LC and SR performed statistical analysis. All authors collected data of patients in their center, contributed to drafting of the work, revised it critically for important intellectual content, and approved the final version of the manuscript. All authors give their agreement to be accountable for all aspects of the work and ensure the accuracy and integrity of any part of the work.

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Data availability

Data availabile upon reasonable request.

Declarations

Conflicts of interest

All authors declare that they have no conflicts of interest related to the manuscript.

Ethical approval

This study was approved by the independent institutional review board of Poitiers, and registered at http://www.clinicaltrials.gov (NCT07022028). In compliance with the national legislation regarding observational retrospective studies, this study was declared to European General Data Protection Regulation (MR004). An information letter was sent to all patients eligible in the study, and patients could be included on the condition that they did not object.

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